

ProtoQSAR API Guide

Beginner-friendly instructions for test and experimental-data access

ProtoQSAR guide: reading CheMatSustain test data

Beginner-friendly instructions for test and experimental-data access

This guide is for ProtoQSAR users. It starts from the beginning and assumes you have not used an API before.

1. What you receive

Each ProtoQSAR user receives their own two credential values:

```
client_id:      cms_...  
client_secret:  ...
```

These values are different from your website email and password.

- The **client ID** identifies your API credential.
- The **client secret** proves that you are allowed to use it.
- Treat the client secret like a password.
- Do not share credentials with another user. Every ProtoQSAR user has their own.
- Never put the secret in source code, Git, screenshots, email, or support tickets.

Your credential can read:

- the complete test catalogue;
- experimental data for every test.

It does not grant protocol or shared-file access.

You can also use the browser-based API Explorer without signing into the main website:

```
https://database.eurskem.com/api-explorer
```

The Explorer still asks for your issued API client ID and client secret. It keeps them only in the current page's memory and removes them when the page is refreshed or closed.

Accessing the new API Explorer UI

The API Explorer is the easiest option when you want to inspect or download data without writing code:

1. Open a current version of Chrome, Edge, Firefox, or Safari.
2. Go to <https://database.eurskem.com/api-explorer>. You do not need to sign into the main CheMatSustain website first.
3. In the **API credential** card, enter your issued **Client ID** and **Client secret**. These are not your website email address and password.
4. Select **Test credentials**. A green **Credential accepted** message confirms that the values work. If you receive 401 Unauthorized, copy both values again and make sure there are no leading or trailing spaces.
5. Select **Run index API** with **All test names** selected and the Test ID empty. This loads the complete live lightweight index and fills the test-name selector with the names currently in the database.
6. To filter by name, choose a value such as MTT and select **Run index API** again. Name matching is exact but is not case-sensitive.
7. To look up a numeric ID, enter it in **Test ID** and select **Run index API**. You may combine a test name and ID; both conditions must match.
8. Select **Open** on a result row, or enter an ID and select **Open test ID**, to retrieve the complete record for that one test.
9. Use **JSON** to download the index or complete record. In the complete-record panel, **Copy** places the displayed JSON on the clipboard.
10. In **Python test**, use **Copy** or **Download .py** to obtain a script based on the selected filters. The generated script reads credentials from environment variables and never contains the secret entered in the page.

Select the bin icon to clear the credentials and results immediately. Closing or refreshing the tab also removes the credentials. The browser does not put them in cookies, browser storage, URLs, generated scripts, or downloaded JSON.

Do not use the Explorer on a shared or public computer. Do not take a screenshot while the client secret is visible.

2. The API address

The base address is:

```
https://database.eurskem.com/api/v1
```

An **endpoint** is a path added to this address. For example, the tests endpoint is:

```
https://database.eurskem.com/api/v1/tests
```

All examples below use HTTPS. Do not change it to HTTP.

3. Easiest first test: curl

curl is a small command-line tool for making web requests. It is already installed on macOS and most Linux systems. On current Windows versions, use it in PowerShell or Windows Terminal.

macOS or Linux

Open Terminal and enter these commands. The second command hides the secret as you type, and neither credential value is written literally into shell history:

```
read -r -p 'Client ID: ' CHEMAT_CLIENT_ID
read -r -s -p 'Client secret: ' CHEMAT_CLIENT_SECRET; echo
export CHEMAT_CLIENT_ID CHEMAT_CLIENT_SECRET
```

Paste each issued value at its prompt and press Enter. Nothing is displayed while you paste the client secret.

Check your credential:

```
curl --fail-with-body --user "$CHEMAT_CLIENT_ID:$CHEMAT_CLIENT_SECRET" \
'https://database.eurskem.com/api/v1/portal/me'
```

Then request the complete lightweight live test index:

```
curl --fail-with-body --user "$CHEMAT_CLIENT_ID:$CHEMAT_CLIENT_SECRET" \
'https://database.eurskem.com/api/v1/test-index'
```

You should receive JSON: structured text beginning with [for a list or { for one object. A successful request has HTTP status 200.

When finished, remove the values from the current terminal session:

```
unset CHEMAT_CLIENT_ID CHEMAT_CLIENT_SECRET
```

Windows PowerShell

Open PowerShell and enter:

```
$env:CHEMAT_CLIENT_ID = 'cms_your_client_id_here'
$env:CHEMAT_CLIENT_SECRET = 'your_client_secret_here'
```

Check your credential:

```
curl.exe --fail-with-body --user "$($env:CHEMAT_CLIENT_ID): $($env:CHEMAT_CLIENT_SECRET)" `
'https://database.eurskem.com/api/v1/portal/me'
```

Request the complete lightweight live test index:

```
curl.exe --fail-with-body --user "$($env:CHEMAT_CLIENT_ID): $($env:CHEMAT_CLIENT_SECRET)" `
  "https://database.eurskem.com/api/v1/test-index"
```

When finished:

```
Remove-Item Env:CHEMAT_CLIENT_ID
Remove-Item Env:CHEMAT_CLIENT_SECRET
```

Production backend terminal (administrators only)

This subsection is only for an authorised administrator who already has shell access to the CheMatSustain production host. ProtoQSAR users should use the public HTTPS commands above; they do not need server access.

On the production host, change to the deployment directory:

```
cd /home/chematsustain
```

Read the client ID normally and the client secret without displaying it on the screen. Neither value is written literally into the shell history:

```
read -r -p 'Client ID: ' CHEMAT_CLIENT_ID
read -r -s -p 'Client secret: ' CHEMAT_CLIENT_SECRET; echo
export CHEMAT_CLIENT_ID CHEMAT_CLIENT_SECRET
```

The hardened backend container does not include `curl`. Use its installed Python `requests` library to check the credential against the backend process inside that same container:

```
docker compose exec -T \
  -e CHEMAT_CLIENT_ID -e CHEMAT_CLIENT_SECRET \
  backend python -c '
import os
import requests

response = requests.get(
    "http://127.0.0.1:8000/v1/portal/me",
    auth=(os.environ["CHEMAT_CLIENT_ID"], os.environ["CHEMAT_CLIENT_SECRET"]),
    timeout=30,
)
response.raise_for_status()
print(response.json())
'
```

Retrieve the unlimited lightweight index from the backend terminal:

```
docker compose exec -T \
  -e CHEMAT_CLIENT_ID -e CHEMAT_CLIENT_SECRET \
  backend python -c '
import os
import requests

response = requests.get(
    "http://127.0.0.1:8000/v1/test-index",
    auth=(os.environ["CHEMAT_CLIENT_ID"], os.environ["CHEMAT_CLIENT_SECRET"]),
    timeout=30,
)
response.raise_for_status()
tests = response.json()
print(f"Returned {len(tests)} tests")
print(tests[:3])
'
```

The `http://127.0.0.1:8000` address is appropriate only inside the backend container because the traffic never leaves that container. All partner and internet access must continue to use `https://database.eurskem.com/api/v1`; never expose port 8000 publicly.

When finished, remove the credential from the host shell:

```
unset CHEMAT_CLIENT_ID CHEMAT_CLIENT_SECRET
```

4. Available methods

All ProtoQSAR requests use the HTTP method GET. GET means “read”; these requests do not create, update, or delete test records.

Method and endpoint	Purpose
GET /portal/me	Check that the credential works
GET /test-index	Return every accessible test's ID, name, Work Package and identifier; no pagination limit
GET /test-index?test_name=MTT	Return every accessible test with one test name
GET /test-index?test_id=3	Look up the lightweight identity fields for one test ID
GET /tests?limit=25&offset=0	Read one page of test summaries
GET /tests/{test_id}	Read the complete data and identity fields for one test
GET /experimental-data/{test_id}	Read detailed data for one test

Replace {test_id} with the numeric id returned by /tests. For example:

```
curl --fail-with-body --user "$CHEMAT_CLIENT_ID:$CHEMAT_CLIENT_SECRET" \
'https://database.eurskem.com/api/v1/experimental-data/3'
```

For new integrations, use the complete single-test endpoint because it includes the ID, test name, Work Package and identifier alongside the data:

```
curl --fail-with-body --user "$CHEMAT_CLIENT_ID:$CHEMAT_CLIENT_SECRET" \
'https://database.eurskem.com/api/v1/tests/3'
```

Unlimited live index

Unlike /tests, the lightweight /test-index endpoint has no pagination limit. It is the authoritative live list and automatically includes tests added after this PDF was generated:

```
curl --fail-with-body --user "$CHEMAT_CLIENT_ID:$CHEMAT_CLIENT_SECRET" \
'https://database.eurskem.com/api/v1/test-index'
```

Filter it by test name:

```
curl --fail-with-body --user "$CHEMAT_CLIENT_ID:$CHEMAT_CLIENT_SECRET" \
'https://database.eurskem.com/api/v1/test-index?test_name=MTT'
```

Each index item contains:

```
{
  "test_id": 3,
  "test_name": "MTT",
  "work_package": "WP3",
  "identifier": "CMS_4a_AuNP"
}
```

5. Understanding the test response

GET /tests returns a JSON list. Each item can contain:

Field	Meaning
id	Numeric identifier used by the experimental-data endpoint
organisation_id	Owning organisation, if assigned; legacy records may be null
work_package_name	CheMatSustain work package
element_cms_id	Consortium element/reference identifier
test_name	Name or type of test
test_details	Summary/details of the test
final_results	Final result summary, if present
statistical_analysis	Statistical analysis, if present
created_at	Creation time
updated_at	Last update time

GET /experimental-data/{test_id} can additionally return raw_data and processed_data. A field may be null when no value has been entered.

6. Pagination: obtaining every test

The API returns at most 25 tests per request. This is called **pagination**.

- limit=25 asks for 25 records.
- offset=0 starts at the first record.
- The next page uses offset=25, then 50, 75, and so on.
- Stop when a page contains fewer than 25 records.

Example second page:

```
curl --fail-with-body --user "$CHEMAT_CLIENT_ID:$CHEMAT_CLIENT_SECRET" \
  'https://database.eurskem.com/api/v1/tests?limit=25&offset=25'
```

Do not set limit above 25; the API will reject it.

7. Python: download the unlimited live test index

Install Python 3, then install the requests package once:

```
python3 -m pip install requests
```

Create a file named `download_test_index.py` with this complete program:

```
import json
import os

import requests

BASE_URL = "https://database.eurskem.com/api/v1"
CLIENT_ID = os.environ["CHEMAT_CLIENT_ID"]
CLIENT_SECRET = os.environ["CHEMAT_CLIENT_SECRET"]

response = requests.get(
    f"{BASE_URL}/test-index",
    auth=(CLIENT_ID, CLIENT_SECRET),
    timeout=30,
)
response.raise_for_status()
tests = response.json()

with open("chematsustain_test_index.json", "w", encoding="utf-8") as output:
    json.dump(tests, output, indent=2, ensure_ascii=False)

print(f"Saved {len(tests)} tests to chematsustain_test_index.json")
```

Set the environment variables as shown in section 3, then run:

```
python3 download_test_index.py
```

The program creates `chematsustain_test_index.json` in the current directory. There is no pagination loop or result limit. It does not write your secret into that output file.

8. Python: retrieve experimental data

This example downloads test 3. Change `TEST_ID` to the test you need:


```

import json
import os

import requests

TEST_ID = 3
response = requests.get(
    f"https://database.eurskem.com/api/v1/experimental-data/{TEST_ID}",
    auth=(os.environ["CHEMAT_CLIENT_ID"], os.environ["CHEMAT_CLIENT_SECRET"]),
    timeout=30,
)
response.raise_for_status()

with open(f"experimental_data_{TEST_ID}.json", "w", encoding="utf-8") as output:
    json.dump(response.json(), output, indent=2, ensure_ascii=False)

```

9. Optional: save basic test fields as CSV

After downloading `chematsustain_test_index.json`, this program creates a spreadsheet- friendly CSV file:

```

import csv
import json

with open("chematsustain_test_index.json", encoding="utf-8") as source:
    tests = json.load(source)

columns = [
    "test_id",
    "test_name",
    "work_package",
    "identifier",
]

with open("chematsustain_tests.csv", "w", newline="", encoding="utf-8-sig") as output:
    writer = csv.DictWriter(output, fieldnames=columns, extrasaction="ignore")
    writer.writeheader()
    writer.writerows(tests)

```

Open the resulting `chematsustain_tests.csv` in Excel or LibreOffice Calc.

10. Postman (no programming)

1. Open Postman and choose **New** → **HTTP Request**.
2. Set the method to **GET**.
3. Enter: `https://database.eurskem.com/api/v1/tests?limit=25&offset=0`
4. Open the **Authorization** tab.

5. Choose **Basic Auth**.
6. Put your `client_id` in **Username**.
7. Put your `client_secret` in **Password**.
8. Click **Send**.
9. Confirm the response status is **200 OK**.

Avoid saving or synchronising a Postman collection containing the real secret. Prefer Postman's local/environment secret variables if you need to reuse it.

11. Common errors

Status or symptom	Meaning	What to do
200 OK	Request worked	Use the returned JSON
401 Unauthorized	Client ID/secret is wrong, disabled, or was copied with extra spaces	Copy both values again; if still failing, contact Eurskem
403 Forbidden	Credential lacks the required scope or resource access	Contact Eurskem; repeatedly retrying will not help
404 Not Found	Test ID/path does not exist, or is not visible	Check the ID came from /tests and check the URL
422 Unprocessable Entity	Invalid pagination, such as limit above 25 or negative offset	Use limit=25 and an offset of 0 or more
500 Internal Server Error	Server-side problem	Record the time and endpoint, then contact Eurskem; do not send your secret
Empty list []	The request worked but returned no records on that page	If paginating, you have reached the end
Request times out	Network/VPN/firewall issue or temporary service issue	Retry once; then send the time and endpoint to support

When requesting support, provide:

- your work email;
- endpoint/path (without credentials);
- HTTP status;
- approximate time and timezone;
- a short description of what you expected.

Never include the client secret in a message or screenshot.

12. Security and credential rotation

- Store the secret in a password manager or approved secret manager.
- Use environment variables for scripts; do not paste the secret into Python.
- Add `.env` files to `.gitignore` before using them.
- Do not disable TLS certificate verification (`verify=False`, `-k`, or `--insecure`).
- Do not share one user's credential among colleagues.
- If a secret is lost or may have been exposed, stop using it and notify Eurskem immediately. Eurskem will disable it and issue a replacement; secrets cannot be recovered from the server.

Support contact: ayush.khandelwal@eurskem.com

13. Worked example: open one test from the list

The appendix below contains every available `test_id` and its associated `test_name`. The same name can appear more than once, so always use the numeric ID from the row you want.

For example, the first row is:

Test ID	Test name
3	MTT

To open the experimental data for this exact test, replace `{test_id}` with 3.

With curl (macOS or Linux)

After setting `CHEMAT_CLIENT_ID` and `CHEMAT_CLIENT_SECRET` as shown in section 3:

```
curl --fail-with-body --user "$CHEMAT_CLIENT_ID:$CHEMAT_CLIENT_SECRET" \
'https://database.eurskem.com/api/v1/experimental-data/3'
```

With Postman

1. Create a new **GET** request.
2. Enter: `https://database.eurskem.com/api/v1/experimental-data/3`
3. Open **Authorization** and select **Basic Auth**.
4. Enter your `client_id` as **Username**.
5. Enter your `client_secret` as **Password**.
6. Click **Send**.
7. A successful response says **200 OK** and returns JSON for test ID 3.

The endpoint is not a normal public web page, so pasting it into a browser without API authentication returns 401 Unauthorized. Use curl, Postman, or the Python method from section 8.

14. Complete test ID and test-name list

This list was exported from the production CheMatSustain database. It contains all 327 tests available when this guide was generated. It is a dated reference, not the live source. Use `/api/v1/test-index` or the API Explorer to see tests added later. Test IDs are permanent record identifiers but are not consecutive. Repeated test names are expected.

Test ID	Test name
3	MTT
4	MTT
5	MTT
6	MTT
7	MTT
8	MTT
9	MTT
10	MTT
12	MTT
14	MTT
16	MTT
28	DLS
32	DLS
36	DLS
37	MTT
41	FTIR
42	FTIR
43	FTIR
44	FTIR
45	FTIR
46	FTIR
47	FTIR
48	FTIR
49	FTIR
51	FTIR
52	DLS

Test ID	Test name
53	DLS
54	DLS
55	DLS
58	DLS
59	DLS
60	DLS
61	FTIR
99	HR-STEM
100	HR-STEM
101	HR-STEM
102	HR-STEM
103	HR-STEM
104	HR-STEM
105	HR-STEM
106	HR-STEM
107	HR-STEM
108	HR-STEM
109	UV-VIS
110	UV-VIS
111	ZETA
121	SIMS
122	SIMS
123	SIMS
124	SIMS
125	SIMS
126	SIMS

Test ID	Test name
127	SIMS
128	SIMS
129	SIMS
130	SIMS
145	XPS
146	UPS
151	XRD
153	UV-VIS
154	ZETA
155	UV-VIS
156	ZETA
157	UV-VIS
158	UV-VIS
159	ZETA
160	UV-VIS
161	XRD
162	UV-VIS
163	UV-VIS
164	XRD
165	UV-VIS
166	XRD
167	UV-VIS
168	XRD
169	UV-VIS
170	XRD
171	UV-VIS

Test ID	Test name
172	UV-VIS
173	UV-VIS
174	UV-VIS
177	UV-VIS
180	UV-VIS
183	UV-VIS
186	SIMS
187	ZETA
188	ZETA
189	ZETA
190	ZETA
191	ZETA
192	ZETA
193	ZETA
195	ZETA
196	ZETA
197	ZETA
198	FTIR
199	HR-STEM
200	SIMS
201	UV-VIS
202	DLS
203	FTIR
204	HR-STEM
205	SIMS
206	UV-VIS

Test ID	Test name
207	DLS
208	FTIR
209	HR-STEM
210	UV-VIS
211	ZETA
212	SIMS
213	DLS
214	FTIR
215	HR-STEM
216	UV-VIS
217	ZETA
218	SIMS
219	FTIR
220	DLS
221	HR-STEM
222	UV-VIS
223	ZETA
224	SIMS
225	DLS
226	FTIR
227	HR-STEM
228	UV-VIS
229	ZETA
230	SIMS
241	TB
242	DLS

Test ID	Test name
243	TGA
244	TGA
245	TGA
246	TGA
247	TGA
248	DSC
249	DSC
250	DSC
251	DSC
252	DSC
254	FTIR
255	FTIR
256	FTIR
257	ZETA
268	ROS
269	ROS
270	ROS
271	ROS
272	ROS
273	ROS
274	ROS
275	ROS
277	ROS
278	ROS
279	ROS
280	ROS

Test ID	Test name
281	ROS
282	ROS
284	ROS
285	ROS
286	ROS
287	ROS
288	ROS
289	ROS
290	ROS
294	TB
295	TB
297	TB
298	TB
299	TB
300	TB
301	TB
302	TB
303	TB
306	TB
307	TB
308	DLS
309	HR-STEM
310	DLS
311	FTIR
312	SIMS
313	FTIR

Test ID	Test name
314	FTIR
315	FTIR
316	FTIR
317	FTIR
318	ZETA
319	HR-STEM
320	DLS
321	UPS
328	UPS
329	UPS
330	UPS
331	UPS
332	UPS
333	UPS
334	XPS
335	XPS
336	XPS
337	XPS
338	XPS
339	XPS
340	XPS
341	MNT
342	MNT
343	ROS
345	ROS
346	FTIR

Test ID	Test name
347	FTIR
348	SIMS
349	SIMS
360	MTT
362	MTT
363	MTT
365	TB-Microfluidic
366	Rotifier
367	Rotifier
368	Rotifier
370	Rotifier
371	TB-Microfluidic
372	TB-Microfluidic
373	TB-Microfluidic
374	TB-Microfluidic
375	TB-Microfluidic
376	TB-Microfluidic
377	TB-Microfluidic
378	TB-Microfluidic
379	TB-Microfluidic
380	TB-Microfluidic
381	TB-Microfluidic
382	TB-Microfluidic
383	TB-Microfluidic
384	TB-Microfluidic
385	TB-Microfluidic

Test ID	Test name
386	TB-Microfluidic
387	TB-Microfluidic
388	TB
389	Rotifier
390	Rotifier
391	Rotifier
392	Rotifier
393	Rotifier
394	Rotifier
395	Rotifier
396	Rotifier
397	Rotifier
398	Rotifier
399	Rotifier
400	Rotifier
401	Rotifier
402	Rotifier
403	Rotifier
404	Rotifier
405	Rotifier
406	Rotifier
407	Rotifier
408	Rotifier
409	TB
411	MTT
412	MTT

Test ID	Test name
413	MTT
414	SIMS
415	SIMS
416	SIMS
417	DSC
419	DSC
420	DSC
421	TGA
422	TGA
423	TGA
424	ROS
426	MTT
427	ROS
428	MTT
430	MTT
432	WaterFlea
433	WaterFlea
434	WaterFlea
435	WaterFlea
436	WaterFlea
437	WaterFlea
438	WaterFlea
439	WaterFlea
440	WaterFlea
441	WaterFlea
442	WaterFlea

Test ID	Test name
443	WaterFlea
444	WaterFlea
445	WaterFlea
446	WaterFlea
447	WaterFlea
448	WaterFlea
449	WaterFlea
450	WaterFlea
451	WaterFlea
452	WaterFlea
453	WaterFlea
454	WaterFlea
455	WaterFlea
456	WaterFlea
457	WaterFlea
458	WaterFlea
459	WaterFlea
460	Algae
461	Algae
462	Algae
463	Algae
464	Algae
465	Algae
466	Algae
467	Algae
468	Algae

Test ID	Test name
469	Algae
470	Algae
471	Algae
472	Algae
473	Algae
474	Algae
475	Algae
476	Algae
477	Algae
478	Algae
479	Algae
480	Algae
481	Algae
482	Algae
483	Algae